

MAT 1272 Statistics
Chapter 4: Probability Rules and Counting
Sections 4.4, 4.5, and 4.6

Learning Objectives

Goals

Students should be able to:

- Use the multiplication rule.
- Distinguish independent and dependent events.
- Use the addition rule.
- Apply the counting rule.
- Evaluate factorials.
- Compute combinations.
- Compute permutations.
- Use the TI-84 calculator for factorials, combinations, and permutations.

4.4 Intersection of Events and the Multiplication Rule

Intersection of Events

The intersection of two events consists of outcomes common to both events.

Notation:

$$A \cap B$$

Read as:

$$A \text{ and } B$$

Joint Probability

The probability

$$P(A \cap B)$$

is called a joint probability.

Independent Events

Multiplication Rule for Independent Events

If events A and B are independent,

$$P(A \cap B) = P(A)P(B)$$

Example

Find the probability of obtaining two heads.

$$P(H) = \frac{1}{2}$$

$$P(HH) = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$$

$$\boxed{\frac{1}{4}}$$

Dependent Events

General Multiplication Rule

If events are dependent,

$$P(A \cap B) = P(A)P(B|A)$$

Example

A box contains 7 red and 13 blue marbles.

Two marbles are selected without replacement.

Find the probability that both are red.

$$P(\text{first red}) = \frac{7}{20}$$

$$P(\text{second red}|\text{first red}) = \frac{6}{19}$$

$$P(\text{both}) = \frac{7}{20} \cdot \frac{6}{19} = 0.1105$$

$$\boxed{0.1105}$$

4.5 Union of Events and the Addition Rule

Union of Events

The union of two events consists of outcomes belonging to event A , event B , or both.
Notation:

$$A \cup B$$

Read as:

$$A \text{ or } B$$

Addition Rule

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Example

Suppose

$$P(A) = 0.44$$

$$P(B) = 0.24$$

$$P(A \cap B) = 0.08$$

Find

$$P(A \cup B)$$

$$= 0.44 + 0.24 - 0.08 = 0.60$$

$$\boxed{0.60}$$

Why Do We Subtract?

When we add

$$P(A) + P(B),$$

the overlap

$$P(A \cap B)$$

is counted twice.

Therefore, we subtract it once.

4.6 Counting Rule

Fundamental Counting Principle

If a process has

$$m$$

ways for Step 1 and

$$n$$

ways for Step 2,
then the process can be completed in

$$m \times n$$

ways.

Example

A restaurant offers:

- 4 appetizers
- 5 entrees
- 3 desserts

Number of meals:

$$4 \times 5 \times 3 = 60$$

$$\boxed{60}$$

Factorials

Factorial

For a positive integer n ,

$$n! = n(n-1)(n-2) \cdots 2(1)$$

Also,

$$0! = 1$$

Examples

$$5! = 120$$

$$7! = 5040$$

$$0! = 1$$

Combinations

Combination

A combination is a selection in which order does NOT matter.

Examples:

- Committees
- Selecting students

Combination Formula

$${}_nC_r = \frac{n!}{r!(n-r)!}$$

Example

Choose 2 flavors from 6.

$${}_6C_2 = \frac{6!}{2!4!} = 15$$

15

Permutations

Permutation

A permutation is a selection in which order DOES matter.

Examples:

- Officers
- Race finishes
- Passwords

Permutation Formula

$${}_nP_r = \frac{n!}{(n-r)!}$$

Example

Select 3 officers from 20 members.

$$\begin{aligned} {}_{20}P_3 &= \frac{20!}{17!} \\ &= 20 \cdot 19 \cdot 18 = 6840 \end{aligned}$$

6840

TI-84 Calculator

Factorials

To compute

5!

1. Type 5
2. Press MATH
3. Select PRB
4. Select 4:!
5. Press ENTER

Combinations

To compute

$${}_6C_2$$

1. Type 6
2. Press MATH
3. Select PRB
4. Select 3:nCr
5. Type 2
6. Press ENTER

Answer:

$$15$$

Permutations

To compute

$${}_{20}P_3$$

1. Type 20
2. Press MATH
3. Select PRB
4. Select 2:nPr
5. Type 3
6. Press ENTER

Answer:

$$6840$$

Group Activity

Work with Your Group

1. Two coins are tossed. Find

$$P(HH).$$

2. If

$$P(A) = 0.4$$

and

$$P(B) = 0.5,$$

find

$$P(A \cap B)$$

assuming the events are independent.

3. If

$$P(A) = 0.6, \quad P(B) = 0.5, \quad P(A \cap B) = 0.2,$$

find

$$P(A \cup B).$$

4. Evaluate

$$6!$$

5. Compute

$${}_8C_3.$$

6. Compute

$${}_8P_3.$$

Summary

Key Formulas

Conditional Probability:

$$P(B|A) = \frac{P(A \cap B)}{P(A)}$$

General Multiplication Rule:

$$P(A \cap B) = P(A)P(B|A)$$

Independent Events:

$$P(A \cap B) = P(A)P(B)$$

Addition Rule:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Combinations:

$${}_nC_r = \frac{n!}{r!(n-r)!}$$

Permutations:

$${}_nP_r = \frac{n!}{(n-r)!}$$